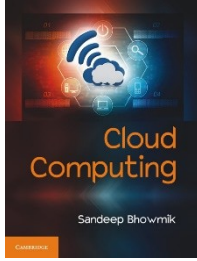


Cloud Computing

Chapter 10

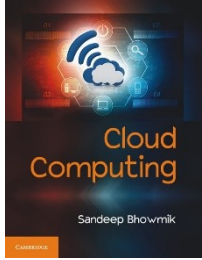
Capacity Planning

Cambridge University Press



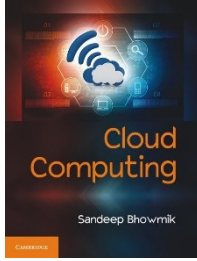
Capacity Planning

- One alluring concept about cloud computing service is its ability of delivering infinite resources.
- But, at the physical (data center) level, it is never possible for to arrange unlimited volume of computing resources.
- Service providers actually create impression of unlimited resources.



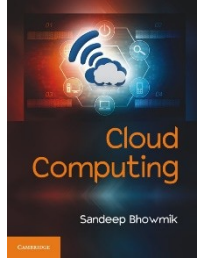
Capacity Planning in Computing

- Capacity planning in computing is a process that –
 - determines the future requirement of computing resources
 - to provide desired levels of service to a given workload at the least cost.



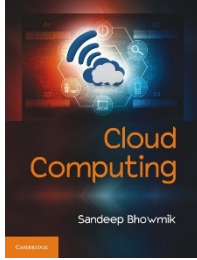
Traditional Way of Capacity Planning

- Traditionally, capacity planning in computing has been performed in silos (isolated storages or data) or in isolated organizational structures.
- The fixed resource capacity within a silo makes it difficult to manage varying resource needs of application with time.
- Moreover, the actual stakeholders had very little control over the capacity planning process.
- Hence possibilities of occurring gaps between actual capacity and actual demand were high.



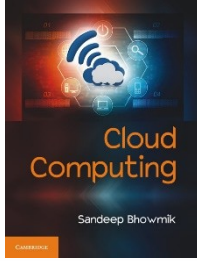
Capacity Planning in Cloud Computing

- The impression of infinite resource availability can only be created through proper capacity planning.
- This is possible with the business model where -
 - one service provider have many clients and
 - enormous number of applications to support from their single or multiple connected data center(s).
- The consumers must provide a rough idea in advance to the service provider regarding varying resource requirements throughout the year.



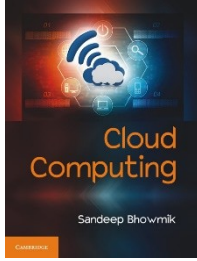
Who Does The Capacity Planning in Cloud

- In cloud computing, capacity planning is done at two levels –
 - At the first level, each consumer of cloud service does its own capacity planning.
 - At the second level, cloud service provider analyzes the capacity requirements of all its consumers together and makes the ultimate overall capacity planning.



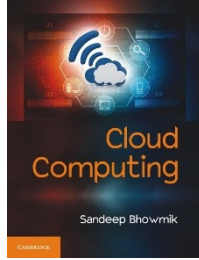
Who Does The Capacity Planning in Cloud

- In this regard, cloud service consumers can be divided into two groups –
 - The IaaS consumers generally plan and reserve a fixed capacity of resources they will consume for a period.
 - The SaaS and PaaS consumers generally opt for dynamically metered resource use. Hence their declaration about service requirements noted in the service level contracts (SLA contract) is considered very important.



Who Does The Capacity Planning in Cloud

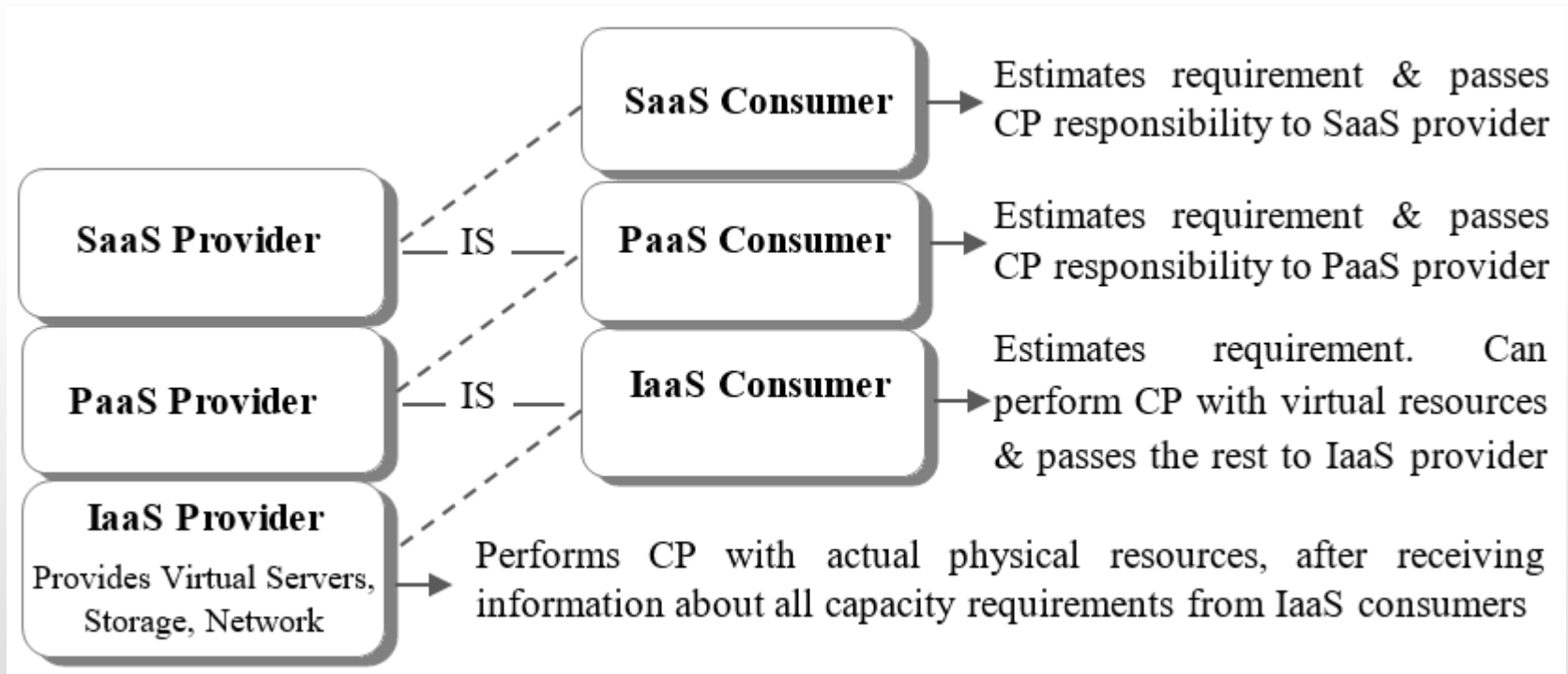
- At the end, the bottom level capacity planning is the responsibility of IaaS provider.
- IaaS provider studies all those capacity requirements along with SLA contracts to know about all consumers' service requirements.
- They perform the actual physical capacity planning for the whole system.

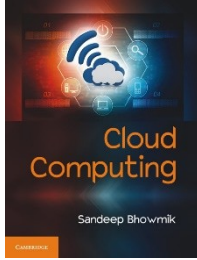


Capacity Planning at Different Service Levels

- The SaaS consumers can pass the capacity planning task to SaaS provider through SLA.
- The SaaS provider in-turn can plan and inform about capacity requirements to the PaaS provider through SLA.
- The PaaS provider as an IaaS consumer,
 - either can perform capacity planning themselves with the virtual resources supplied by IaaS provider,
 - or, can again pass the capacity planning task to IaaS provider through a well-planned SAL agreement.

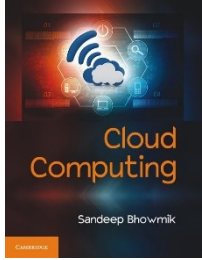
Capacity Planning at Different Service Levels





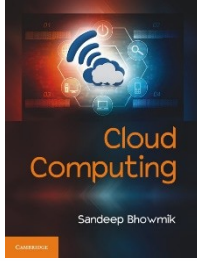
Role of Service-Level Agreement

- Service Level Agreement (SLA) contains the details of the contract made between the service provider and consumer.
- Consumers may declare the resource demands in the SLA to ensure availability.
- Apart from predictable average requirement, this should include mention of the periods when demand grows.
- Thus, the SLAs play important role in the capacity planning process in cloud computing.



Capacity Planning – Then and Now

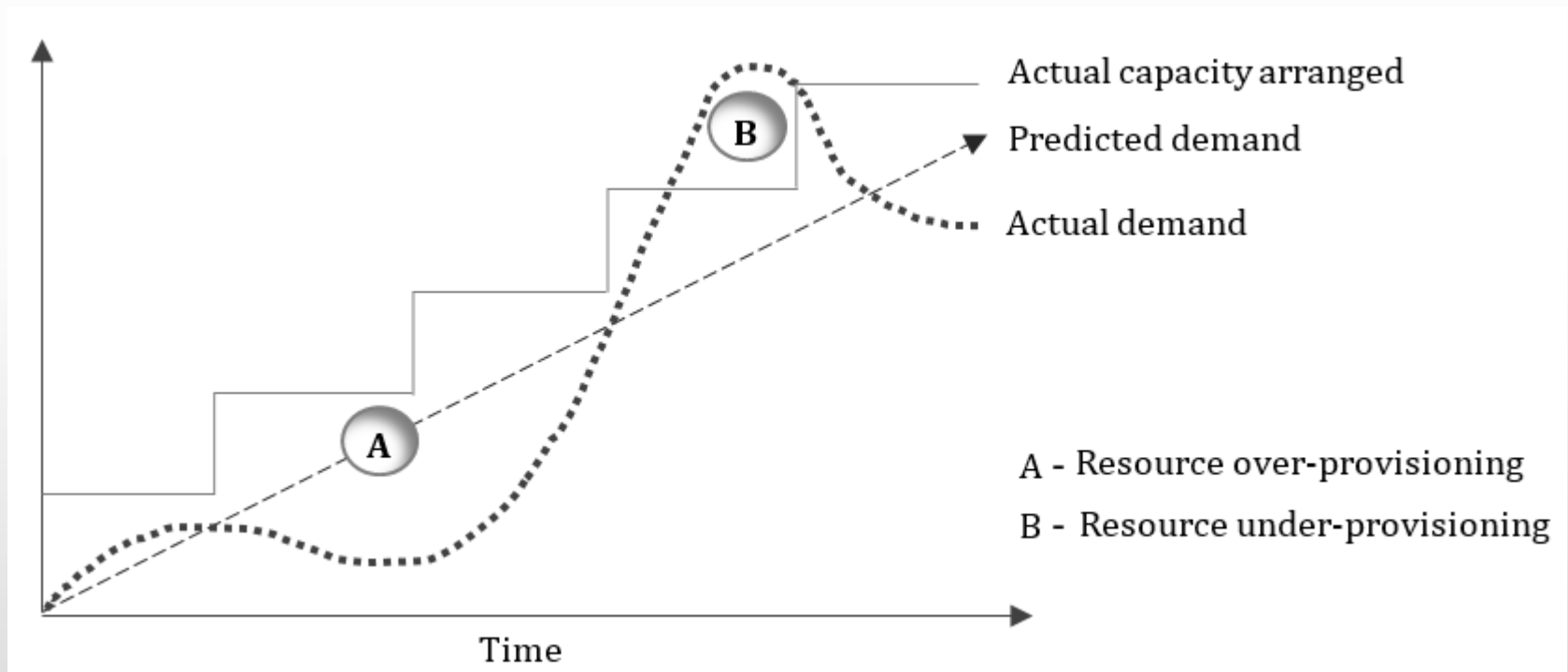
- Traditional Fixed Cost Computing Model
- Modern Variable Cost Computing Model

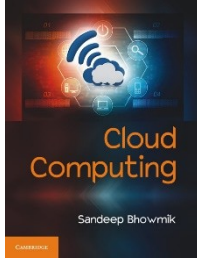


Traditional Fixed Cost Computing Model

- Traditionally computing capacity planning of organizations was based on fixed cost model.
- In that approach, there were possibilities of occurrence of two different problematic scenarios –
 - First one due to over-provisioning of resources.
 - The other one due to under-provisioning of resources.

Traditional Fixed Cost Computing Model

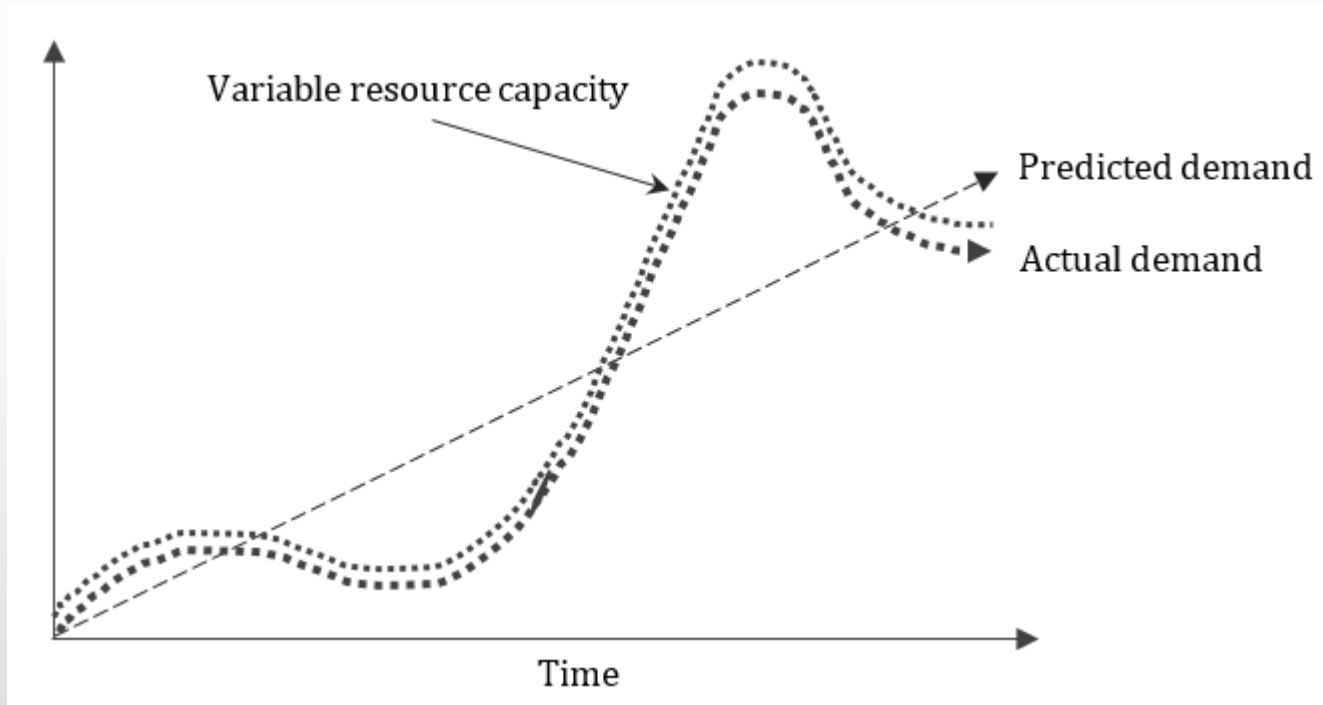


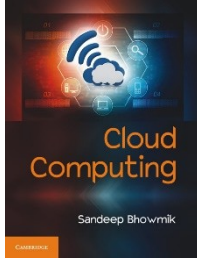


Modern Variable Cost Computing Model

- Consumers under this model rarely acquire any unnecessary volume of permanent computing resources, and use resources on rent.
- This model can closely match the resource capacity with actual demand and thus decreases cost of computing by reducing resource wastage.

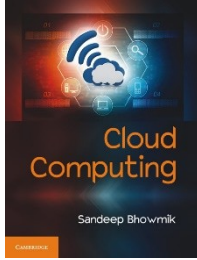
Modern Variable Cost Computing Model





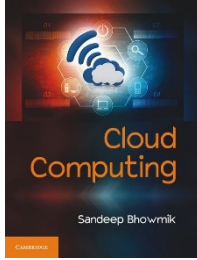
Approaches for Maintaining Sufficient Capacity

- **Traditional Approach with N+1 Rule**
- This approach suggests to maintain one additional resource node than actually required.
- ‘+1’ is the additional capacity added in case demand exceeds supply of ‘N’.
- So, in case one node fails or gets interrupted, that won’t have any impact to the application



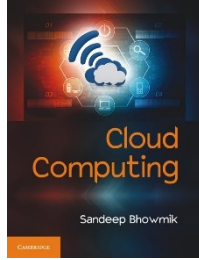
Approaches for Maintaining Sufficient Capacity

- **Traditional Approach with N+1 Rule**
- This approach has obvious limitations.
- Depending on the business requirements, the N+1 rule may not always be sufficient.



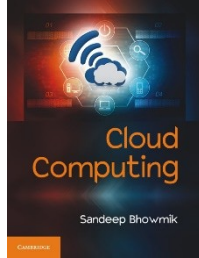
Approaches for Maintaining Sufficient Capacity

- **Cloud Specific Approach**
- Cloud infrastructure service with virtual resources offer a more flexible and scalable approach.
- Here, auto-scaling mechanism always maintains sufficient resource capacity for applications to serve legitimate loads.
- Auto-scaling relieves the cloud service consumers from the tedious job of monitoring and maintaining sufficient capacity.



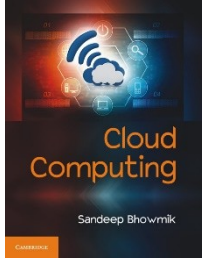
Role of auto-scaling in Capacity Planning

- The traditional approach was to maintain infrastructure keeping the peak capacity of each application in the mind.
- This results in wastage of resources, as most part of the resources would remain unutilized for most of the time.
- Auto-scaling facilitates little or no impact to the applications running on infrastructure.
- Moreover, resources not being used by some application are released to be added in the resource pool.



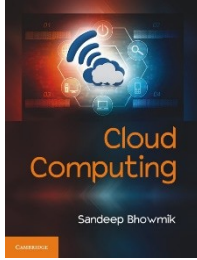
Capacity and Performance

- System capacity and system performance are not same.
- Performance enhancement of any system is performed through system optimization or performance tuning.
- Appropriate capacity planning undoubtedly improves system performance, but the main objective of capacity planning is to meet the future demand of workloads.
- With capacity, the concern is about how much work a system can do, whereas with performance, concern is the rate at which work gets done.



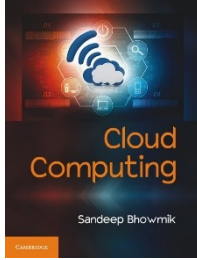
Steps for Capacity Planning

- There are three basic steps for capacity planning -
 - Determining expected demand
 - Analyzing current response to load
 - Knowing the value of system



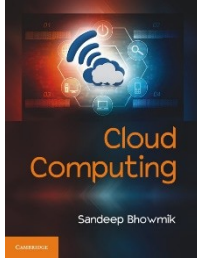
Steps for Capacity Planning

- **Determining expected demand**
- Capacity planners need to determine individual depends of all the applications supported by the system.
- They can aggregate those results to get idea about traffic pattern of the whole system.
- Capacity planners should also evaluate the weekly and yearly statistics on an ongoing basis.



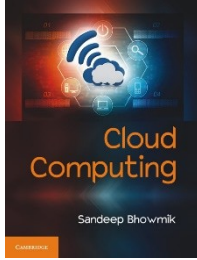
Steps for Capacity Planning

- **Analyzing Current Load**
- System load can be analyzed by measuring the load of different system resources.
- Several system metrics are used to represent these loads. They are –
 - processor
 - memory
 - storage
 - network connectivity



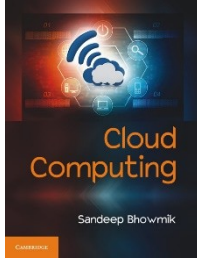
Steps for Capacity Planning

- **Analyzing Current Load**
- Hence the major system parameters are –
 - processor speed
 - memory access speed
 - disk I/O access speed
 - network I/O access speed
- Loads on each of these resources can be measured by operating-system-specific tools.



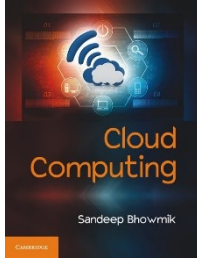
Steps for Capacity Planning

- **Analyzing Current Load**
 - Each of the resources has a utilization rate.
 - When a resource reaches the ceiling, that limits performance of the system. This limit is called *stress point*.
 - To identify the stress point of a resource it is necessary to create corresponding resource utilization curve.



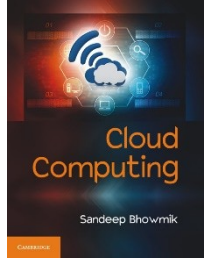
Steps for Capacity Planning

- **Analyzing Current Load**
- The resource utilization statistics can easily identify the bottleneck or the stress point that occurs first.
- Capacity planner must take care of this bottleneck to improve performance of the system, and repeat the load testing activity to find the next bottleneck.



Value of System Capacity

- It is important to know how more capacity can help the business before adding any more resources to the system.
- The decision of capacity enhancement depends on, how much it is going to cost to add capacity against the value of that additional capacity to the business.



Thank You